

Literature Review Report On
Study AI (Generative AI Project)



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Abstract

Generative AI has advanced very quickly and changed many spaces such as education. Existing AI-based education tools are still limited, mostly just using text and do not take full advantage of potential multimodality in education. This project aims to create a holistic AI-based Study Companion by combining Retrieval Augmentation Generation (RAG), voicebased interaction, and multimodal learning methods to improve an individual learner's experience. Once the learner can provide any form of learning material (PDFs, URLs, or topics) to the Study Companion, it produces a structured summary from this material, along with a quiz and an explanation about what was provided to it. The Study Companion also creates images or reads to the user, providing support for a learner whose needs may not have been met by the material previously provided to the user. One of the biggest innovations of the Study Companion is its capability of grounding responses through RAG methods thus, giving an accurate response from the original content and also minimizing the problems found with hallucination when using generative AI.

In addition, it has incorporated a system that can track progress, analyze user performance, identify trouble spots or weaknesses in their performance, and then dynamically adjust the content. Its design includes front-end applications, back-end processing engines, vector databases, and artificial intelligence models in order to provide users with a cohesive learning experience.

The solutions proposed here solve key deficiencies in current systems: absence of personalized expertise; no auditory teacher; limited multimodal integration. These improvements create a much-needed step toward building intelligent tutoring systems for the future.

Introduction

1.1 Background

In the past decade, education has undergone significant changes due to the introduction of Artificial Intelligence (AI). Traditional textbook-based learning methods with conventional, standardized teaching methods are being replaced with more dynamic, technology-based learning environments. Of these advancements, Generative AI has emerged as an effective tool for helping educators produce human-like text, generate images, and help learners interactively.

The use of Large Language Models (LLMs) as well as other AI-based tools including chatbots, has created a substantial increase in students' ability to access information and receive instantaneous explanation of questions. The ability of these systems to answer students' questions, summarize content, and generate practice problems can be compared to tutoring. The current limitations are that many of the existing AI tools available have a limited scope, as they can only be used in a text-based form and cannot fully accommodate many of the various ways individuals learn (e.g., visually and audibly).

Besides being independent tools, existing AI applications are also used as separate learning platforms, with students having to jump back and forth between several applications in order to perform functions such as creating notes, preparing for quizzes, comprehending content, and recording progress with their revisions. These disjointed experiences hinder productivity, thereby decreasing the amount of value students receive from the use of AI.

1.2 Problem Identification

Even though the usage of artificial intelligence (AI) in education has significantly increased in recent times, a substantial gap exists between what generative AI theoretically can do and how it is actually being used in on-the-ground learning environments. Furthermore, the majority of the existing educational systems do not provide a cohesive, adaptable, and multimodal learning experience for students.

One of the major challenges with current educational tools that utilize AI technology is that they are mainly text-based in nature. They can create explanations and respond to questions—however, they tend to do so without the inclusion of multiple types of media (e.g., images, diagrams, verbal explanations). Therefore, students who learn through visual and/or audio means are not gaining all of the information that would be available to them if multiple forms of media were used in classroom settings (e.g., testing).

Another significant limitation associated with many AI-enabled educational tools is that there is no voice-enabled tutoring systems that are integrated into these tools. Listening to an explanation of how to do something is an important aspect of education, especially for students that are trying to learn complex subject areas and listening to someone explain the material makes it easier for the student to retain the information. Many AI-based / AI-assisted educational tools do not offer students either real-time or interactive voice-based assistance therefore; numerous students in a classroom setting must find other resources to help them understand how to learn the material being discussed.

In addition to the lack of reference material, time-consuming and repetitive work related to creating study materials is another major practical challenge that students will face in preparing to study. Manually producing study materials can be cumbersome, whether converting a PDF into structured notes, creating quizzes for practice, or summarizing lengthy content. Current AI systems have limited automation capabilities for automation and provide no complete end-to-end experience.

Personalization of current systems also presents a gap. Very few of the systems available today analyze patterns of learning, or identify areas of weakness. Without this type of data, it will be nearly impossible to provide students with adaptive and personalized learning experiences. This means that students

cannot measure their progress, nor can they identify the focus areas of improvement.

Another factor that plays a significant role in the fragmentation of functions on multiple platforms is the fragmentation of functions. For example, one platform might provide summarization, another might provide quiz generation, and another might provide vocal support. Because these products do not all integrate with each other, users are required to continually switch between tools, causing an inefficient workload and mental confusion.

1.3 Need for the Proposed System

The limitations identified above highlight the urgent need for a **comprehensive and integrated AI-based learning system**. Such a system should not only generate content but also support multiple modes of learning, ensure accuracy, and adapt to individual user requirements.

There is a growing demand for:

- **Multimodal learning systems** that combine text, images, and audio
- **Automated content generation tools** that reduce manual effort
- **Personalized learning platforms** that adapt based on user performance
- **Accurate and reliable AI systems** that minimize hallucination through grounding techniques

An ideal solution should bridge the gap between isolated AI functionalities and a unified learning ecosystem. It should provide a seamless experience where students can input raw learning material and receive structured, interactive, and personalized outputs.

1.4 Objective of the Study

In response to the identified challenges, this project aims to design and develop an **AI-powered Study Companion** that integrates multiple advanced technologies into a single platform.

The primary objective is to build a system that:

- Combines **multimodal learning capabilities** (text, image, and audio)
- Utilizes **Retrieval-Augmented Generation (RAG)** to ensure accurate and context-aware responses
- Provides **voice-based interaction** to enhance accessibility and understanding
- Automates tasks such as **note generation, quiz creation, and content summarization**

By addressing the limitations of existing systems, the proposed solution aims to create a **complete AI-driven learning ecosystem** that enhances efficiency, improves understanding, and supports personalized education.

Literature Review

In recent years, there has been increasing examination of how Generative Artificial Intelligence can be incorporated into education; specifically, how it could be used to improve student engagement, automate the generation of

content and provide tailored learning experiences for students. There have been many systems developed, including chat bots specific to a certain subject, intelligent tutoring systems and multimodal AI systems. Intriguingly enough, even though many advancements have been made regarding the above technologies, they still remain limited in terms of multi-modal integration, scalability and personalization.

The following section will review key research related to AI-powered educational systems covering issues related to methodology, findings and limitations.

5.2 Review of Existing Systems

1) **ChatISA** -Domain-Specific AI Chatbot

An AI chatbot developed for use in the Information Systems domain. ChatISA uses Chatbots as a method of building Domain Specific AI chatbot systems that can follow the DSRM research method and integrate RAG [1]. This is accomplished by providing an AI chatbot which can provide context-aware responses as well as by being able to provide context in a specific way. The research also shows that the use of domain specific chatbots can enhance the student's ability to interact and understand a given subject. However, because ChatISA is limited to a specific subject area, it is not applicable to a wider range of educational contexts, i.e., all of education.

2) **EDUBOT** – AI-Based Student Assistant

EDUBOT utilizes artificial intelligence in the form of an advanced natural language processing system based on language models (e.g., GPT-4) and technologies (e.g., Google Custom Search) [2]. The chatbot leverages external data sources through Application Programming Interfaces (APIs) that provide various Google products and services to allow for near-instantaneous response times for student engagement (and thus increased engagement) with content-related queries; performance visualizations also utilize library tools (e.g., Matplotlib).

As a result of the reliance on external APIs, however, the scalability and reliability of EDUBOT are limited. In addition, EDUBOT currently has limited capabilities in creating multiple modes of input as well.

3) Generative AI in Intelligent Tutoring Systems (ITS)

The recent advancement of Large Language Models (LLMs) in Intelligent Tutoring Systems (ITS) enables automated question generation and adaptive feedback. These systems use students' answers to alter their content in real-time[3]

Although personalized learning and engagement have been greatly enhanced with ITS technology, the majority of ITS applications are still primarily limited to textbased formats with minimal support for multimodal interaction. In addition, concerns about the accuracy and reliability of pedagogical methods remain.

4) Large Multimodal Foundation Models

Multimodal AI combines text, imagery and sometimes audio capabilities to create enhanced interaction experiences through their use of transformer-based architectures typically linked via application programming interfaces (APIs) or middleware systems [4]

The use of multimodal systems creates more engaging learning experiences. However, issues such as hallucination, computational complexity and reliance on human oversight create challenges for the implementation of multimodal systems. As a result, many experiments to improve access to these types of systems are conducted under resource-limited conditions.

Paper	Focus Area	Methadology	Key Findings	Limitation
ChatISA	In-house AI chatbot for IS education	Design Science Research (DSRM), multi-model integration, Retrieval-Augmented Generation (RAG), open-source deployment	Domain-specific chatbot improves student interaction	Limited generalization, domain restriction
EDUBOT	AI-powered student assistant chatbot	GPT-4 integration, Google Custom Search API, performance visualization (Matplotlib), supervised evaluation	Improved engagement & academic performance	API dependency, limited multimodal depth
Generative AI in ITS	AI-enhanced Intelligent Tutoring Systems	Large Language Models (LLMs), automated question generation, adaptive feedback mechanisms	Dynamic question creation & personalized feedback	Pedagogical accuracy concerns, text-dominant interaction
LMFMs in Education	Large Multimodal Foundation Models	Multimodal transformer models, API integration, middleware architecture connecting AI systems	Rich multimodal interaction, enhanced engagement	Digital divide, hallucination risk, need for human oversight

5.3 Key Insights from Literature Review

From the reviewed studies, several important observations can be made:

- Most systems are **chatbot-based and text-oriented**

- Multimodal capabilities are **limited or partially implemented**
- Voice-enabled tutoring is **largely absent**
- Many systems depend heavily on **external APIs**
- **Hallucination and accuracy issues** remain unresolved
- Lack of **integrated progress tracking systems**

These findings clearly indicate that current solutions are fragmented and fail to provide a complete learning ecosystem.

Methodology

The design of the AI Study Companion is modular and incorporates a pipeline method; it converts raw learning input into output that is structured, personalised and multimodal. Some of the more advanced methodologies can include Retrieval-Augmented Generation (RAG), prompt engineering, multimodal content generation and analytics of user performance.

Each of the stages represented by different types of modules can be interrelated; each module has a certain function but all of them contribute to the overall learning experience. Therefore, the completed outputs that will be generated

will provide an accurate representation of the user's learning style and will meet the individual needs of the user.

9.1 System Workflow

The overall workflow of the system consists of the following stages:

1. User Input Acquisition
2. Data Processing and Preparation
3. RAG-based Retrieval
4. Prompt Engineering Layer
5. Output Generation
6. Multimodal Enhancement

Each stage is explained in detail below.

Step 1: User Input Acquisition

The system is designed to accept multiple forms of input to ensure flexibility and usability. Users can provide learning material through:

- Uploading PDF documents
- Entering URLs (web-based content)
- Directly typing topics or questions

This flexibility allows the system to cater to different learning scenarios, whether the user is studying from textbooks, online resources, or conceptual topics.

Once the input is received, it is forwarded to the processing module for further analysis.

Step 2: Data Processing and Preparation

This stage focuses on converting raw input into a structured format suitable for AI processing. **Key Processes:**

- **TextExtraction:** Content is extracted from PDFs or web pages using parsing techniques.

- **TextCleaning:** Removal of unnecessary symbols, formatting issues, and noise.
- **Chunking:** The extracted text is divided into smaller segments (chunks) to improve retrieval accuracy and processing efficiency.
- **EmbeddingGeneration:** Each text chunk is converted into a vector representation using embedding models. These embeddings capture semantic meaning and are essential for similarity-based retrieval.
- **StorageinVectorDatabase:** The generated embeddings are stored in a vector database, enabling efficient retrieval of relevant content during query processing.

This step ensures that the system has a structured and searchable knowledge base derived from user-provided material.

Step 3: Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation is a critical component of the system, responsible for ensuring accuracy and context-awareness.

Working Mechanism:

- When a user query is received, the system searches the vector database for the most relevant text chunks based on semantic similarity.
- These retrieved chunks are then injected into the input prompt as contextual information.
- The language model generates responses based on both the query and the retrieved context.

Importance:

- Reduces hallucination by grounding responses in actual data
- Improves relevance and accuracy
- Ensures alignment with user-provided study material

This approach bridges the gap between generative AI and factual correctness.

Step 4: Prompt Engineering Layer

The prompt engineering layer plays a crucial role in guiding the AI model to generate appropriate outputs.

Features:

- **Role-based Prompting:**
The system assigns specific roles to the AI (e.g., tutor, quiz generator, summarizer) to produce targeted outputs.
- **Task-specific Prompts:**
Different prompts are designed for:
 - Summarization
 - Quiz generation
 - Concept explanation
 - Code generator/explainer
- **Context Injection:**
Retrieved content from the RAG system is embedded within the prompt to ensure relevance.

This layer ensures that the AI produces structured, meaningful, and task-oriented responses.

Step 5: Output Generation

Based on the processed input and engineered prompts, the system generates multiple types of outputs:

- **Structured Summaries:**
Concise and organized notes derived from input material
- **Quizzes and Practice Questions:**
Automatically generated questions to test understanding
- **Detailed Explanations:**
Step-by-step explanations of concepts

The output generation process is dynamic and adapts based on user input and context.

Step 6: Multimodal Enhancement Layer

To support different learning styles, the system incorporates multimodal capabilities.

Components:

- **Image Generation:** Visual representations such as diagrams and concept illustrations are generated using models like Stable Diffusion.
- **Text-to-Speech (TTS):** Generated explanations are converted into audio using libraries such as gTTS, enabling auditory learning.
- **Downloadable Content:**
Users can download notes and generated materials for offline use.

Significance:

- Enhances engagement
- Supports visual and auditory learners
- Makes learning more interactive

This layer transforms the system from a simple chatbot into a comprehensive learning assistant.

EXPERIMENTAL RESULTS

10.1 Overview

The proposed AI Study Companion was evaluated based on its ability to generate structured learning content, provide accurate responses, and support multimodal interaction. Since the system is application-oriented, the evaluation focuses on **functional performance, output quality, and usability** rather than purely numerical benchmarks.

The system was tested using multiple types of inputs such as PDF documents, user-defined topics, and web-based content. Each module was analyzed independently to ensure proper functionality and integration within the overall system.

10.2 Evaluation Criteria

The system was evaluated using the following criteria:

- **Accuracy of Generated Content** The correctness and relevance of summaries, explanations, and answers generated by the model.
 - **Context Awareness (RAG Performance)** Ability of the system to generate responses grounded in user-provided documents.
 - **Output Quality** Clarity, structure, and usefulness of generated notes, quizzes, and explanations.
 - **Multimodal Functionality**
Performance of image generation and text-to-speech modules.
 - **Response Time**
Time taken to process input and generate outputs.
 - **User Experience** Ease of interaction across different modules such as Study Chat, Quizzes, and Voice Chat.
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10.3 Results of Individual Modules

1. Ingest Notes Module

- Successfully processed PDF and text inputs.
- Efficient text extraction and chunking were achieved.
- Embeddings stored in the vector database enabled fast retrieval.

Observation:

The module performed reliably and formed a strong foundation for downstream tasks.

2. Study Chat (RAG-Based Interaction)

- Generated context-aware responses based on uploaded material.
- Reduced hallucination compared to standard LLM responses.
- Provided relevant explanations aligned with input documents.

Observation:

RAG significantly improved response accuracy and trustworthiness.

3. Summarization Module

- Produced concise and structured summaries.
- Maintained key concepts while reducing content length.
- Different summary formats (short and detailed) worked effectively.

Observation:

Helped reduce manual effort in note preparation.

4. Quiz Generation Module

- Generated meaningful and topic-relevant questions.
- Included a mix of conceptual and factual questions.
- Useful for self-assessment and revision.

Observation:

Improved engagement and understanding of concepts.

5. Flashcards Module

- Successfully converted content into quick revision cards.
- Enabled efficient memory recall.

Observation:

Effective for last-minute revision and concept reinforcement.

6. Image Generation Module

- Generated visual representations based on input prompts.
- Useful for conceptual visualization.

Observation:

Enhanced understanding, especially for visual learners.

7. Voice-Based Modules

Voice Chat

- Enabled interaction using speech input.

- Provided conversational responses.

Text-to-Speech

- Converted generated explanations into audio format.

Observation:

Improved accessibility and supported auditory learning.

8. Convert Module

- Successfully handled document format conversions.
- Simplified file handling for users.

10.4 Performance Analysis

- The system demonstrated **high accuracy when RAG was used**, as responses were grounded in user data.
- Response time remained within acceptable limits for most operations.
- Multimodal features increased engagement and usability.

However, certain limitations were observed:

- Performance depends on API availability and internet connectivity
- Image generation may not always produce precise diagrams
- Slight latency in voice processing

10.5 Summary of Results

Overall, the system successfully achieved its objectives by:

- Automating content generation (notes, quizzes, summaries)
- Providing context-aware and accurate responses
- Supporting multimodal learning (text, image, audio)
- Delivering an integrated learning experience

The experimental results confirm that the proposed system is **effective, practical, and scalable for real-world educational use**.

11. RESEARCH GAP

11.1 Limitations in Existing Research

Despite significant advancements in the application of Generative Artificial Intelligence in education, current research and systems exhibit several critical limitations.

Firstly, most existing systems are **predominantly text-based**. While they are capable of generating explanations, summaries, and responses, they do not effectively integrate other learning modalities such as images and audio. This restricts their usability for students who rely on visual or auditory learning methods.

Secondly, many AI-based educational tools are **domain-specific**. Systems like ChatISA are designed for a particular subject area, limiting their applicability across broader academic disciplines. This lack of generalization prevents the development of a universal learning platform.

Another key limitation is the **absence of proper grounding mechanisms** in many systems. Although Large Language Models generate fluent responses, they often suffer from hallucination, producing incorrect or misleading information. Only a limited number of systems incorporate techniques such as Retrieval-Augmented Generation (RAG) to ensure factual accuracy.

Additionally, there is a strong dependence on **external APIs and third-party services**, as seen in systems like EDUBOT. This creates issues related to scalability, reliability, and cost, making such systems less sustainable for largescale deployment.

Furthermore, most existing solutions lack **integration of multiple functionalities**. Features such as summarization, quiz generation, voice interaction, and visualization are often available across different platforms rather than within a single unified system. This fragmentation leads to inefficient workflows and poor user experience.

11.2 Problems Not Yet Solved

Based on the literature review, several important problems remain unresolved in the field of AI-based education systems:

- **Lack of Multimodal Integration:** There is no comprehensive system that effectively combines text, image, and audio-based learning into a single cohesive platform.
 - **Absence of Voice-Enabled Tutoring:** Most systems do not support realtime or interactive voice-based learning, limiting accessibility and engagement.
 - **Inaccurate or Hallucinated Responses:** Many AI models generate responses without grounding them in user-provided content, leading to reliability issues.
 - **Fragmented Learning Tools:** Students are required to use multiple applications for different tasks such as note-making, quizzes, and explanations.
 - **Limited Automation of Study Material Creation:** Existing systems do not fully automate processes like summarization, quiz generation, and content structuring in an integrated manner.
 - **Lack of Practical, Scalable Implementations:** Many research systems remain theoretical or experimental, with limited real-world deployment and usability.
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11.3 Scope for Improvement

The identified gaps highlight several opportunities for improving current AI-based learning systems:

- **Development of Unified Learning Platforms:** There is a need for systems that integrate multiple functionalities such as summarization, quiz generation, voice interaction, and visualization within a single interface.
- **Incorporation of Multimodal Learning:** Future systems should support text, images, and audio to cater to diverse learning styles.
- **Adoption of Retrieval-Augmented Generation (RAG):** Implementing RAG can significantly improve accuracy by grounding responses in relevant data sources.
- **Enhancement of Voice-Based Interaction:** Integrating speech recognition and text-to-speech technologies can improve accessibility and user engagement.
- **Reduction of API Dependency:** Building more self-contained systems can enhance scalability and reliability.
- **Improved User Experience and Automation:** Systems should focus on reducing manual effort by automating repetitive tasks and providing a seamless workflow.

11.4 Summary

The analysis of existing research clearly indicates that current AI-based educational systems are fragmented, limited in modality, and often lack accuracy and integration. Addressing these gaps provides a strong foundation for the development of more advanced, scalable, and user-centric learning platforms.

The proposed AI Study Companion is designed specifically to bridge these gaps by combining multimodal learning, RAG-based accuracy, and integrated functionalities into a unified system.

12. ADVANTAGES AND DISADVANTAGES

12.1 Advantages (Pros)

The proposed AI Study Companion offers several significant benefits over traditional learning methods and existing AI-based educational tools:

- **Multimodal Learning Support**
The system integrates text, image, and audio-based learning, making it suitable for different types of learners, including visual and auditory learners.
- **Automated Content Generation** It eliminates the need for manual note-making by automatically generating summaries, quizzes, and explanations from input material.
- **Context-Aware Responses using RAG**
The use of Retrieval-Augmented Generation ensures that responses are grounded in user-provided data, improving accuracy and reducing hallucination.
- **Unified Learning Platform**
Multiple functionalities such as study chat, quiz generation, summarization, and voice interaction are integrated into a single system, reducing the need to switch between different tools.
- **Time Efficiency** The automation of repetitive tasks such as summarization and question generation significantly reduces the time required for study preparation.
- **Enhanced Engagement** Interactive modules like quizzes, flashcards, and voice chat improve user engagement and make learning more dynamic.

- **Accessibility and Ease of Use** The inclusion of voice-based features allows users to interact with the system hands-free, improving accessibility.
- **Scalability and Flexibility** The modular design allows the system to be extended with additional features in the future.

12.2 Disadvantages (Cons)

Despite its advantages, the system also has certain limitations and challenges:

- **Dependency on Internet and APIs** The system relies on external APIs and internet connectivity, which may affect performance and availability in offline environments.
- **Computational Cost** Running AI models, especially for multimodal features like image generation, requires significant computational resources.
- **Latency in Response Generation** Some modules, particularly voice processing and image generation, may experience slight delays.
- **Accuracy Limitations** Although RAG improves accuracy, there is still a possibility of incomplete or partially incorrect responses in complex scenarios.
- **Limited Image Precision** Generated images may not always represent exact educational diagrams or structured visuals required for certain subjects.
- **Scalability Challenges** As the number of users increases, maintaining performance and response time may become challenging without proper optimization.
- **Security and Data Privacy Concerns** Handling user-uploaded documents and data requires proper security mechanisms to ensure privacy and protection.
- **Initial Setup Complexity** Integration of multiple technologies (LLMs, vector databases, APIs) increases system complexity during development and deployment.

13. CONCLUSION AND FUTURE WORK

13.1 Conclusion

This project presented the design and implementation of a **Multimodal AI-Powered Study Companion** that integrates advanced technologies such as Retrieval-Augmented Generation (RAG), prompt engineering, and multimodal content generation.

The system successfully addresses key limitations in existing AI-based educational tools, including lack of integration, absence of multimodal capabilities, and limited personalization. By combining multiple functionalities into a single platform, the proposed solution provides a seamless and efficient learning experience.

The use of RAG ensures that responses are context-aware and grounded in user-provided material, thereby improving accuracy and reducing hallucination. Additionally, the inclusion of image generation and voice-based interaction enhances engagement and supports diverse learning styles.

The modular architecture of the system allows users to perform multiple learning tasks such as note generation, quiz creation, summarization, and interactive doubt-solving within a unified interface. This significantly reduces the need to switch between different tools, improving productivity and learning efficiency.

Overall, the project demonstrates the practical application of Generative AI in education and highlights its potential in building intelligent tutoring systems.

13.2 Future Work

Although the system performs effectively, there are several areas where further improvements can be made: **1. Real-Time Voice Interaction**

Enhancing the voice module to support real-time conversational interaction with minimal latency. **2. Offline Functionality**

Developing offline capabilities to reduce dependency on internet connectivity and external APIs.

3. Improved Image Generation

Integrating more advanced models for generating accurate educational diagrams and structured visuals.

4. Advanced Personalization

Incorporating deeper learning analytics to provide highly customized learning paths based on user behavior.

5. Integration with Learning Management Systems (LMS)

Connecting the system with platforms such as Moodle or Google Classroom for wider adoption.

6. Reduced Hallucination

Further improving RAG mechanisms and incorporating verification layers to enhance reliability.

7. Mobile Application Development

Developing a mobile version of the system to increase accessibility and usability.

13.3 Final Remark

The proposed AI Study Companion represents a step toward the future of **Aldriven education**, where learning becomes more interactive, personalized, and efficient. With further improvements, such systems have the potential to revolutionize how students learn and interact with educational content.

14. REFERENCES

The following references have been used to study existing systems, methodologies, and technologies related to Generative AI in education and intelligent tutoring systems:

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